THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 6: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For every whole number n, $6n + 3$ is: 1

(a) an odd multiple of 3 **(b)** a multiple of 9
(c) a multiple of 6 **(d)** even

2. If $1 + 2 + 3 + \cdots + n = 666$, then n is: 1

(a) 37 **(b)** 35 **(c)** 38 **(d)** 36

3. The number of reverse-and-add steps for 69 to become a palindrome is: 1

(a) 5 **(b)** 3 **(c)** 4 **(d)** 6

4. The largest 4-digit number with all digits different that is divisible by 3 is: 1

(a) 9876 **(b)** 9878 **(c)** 9875 **(d)** 9877

5. Kaprekar's routine on 4176 reaches 6174 after how many steps? 1

(a) 1 **(b)** 3 **(c)** 0 **(d)** 2

6. The Collatz sequence from 19 needs how many steps to reach 1? 1

(a) 20 **(b)** 22 **(c)** 21 **(d)** 19

7. A two-digit number plus its reverse is always a multiple of: 1

(a) 7 **(b)** 10 **(c)** 11 **(d)** 9

8. If $1 + 2 + 3 + \cdots + n = 1035$, then n is: 1

(a) 47 **(b)** 44 **(c)** 46 **(d)** 45

9. **Assertion (A):** 7 + 8 + 9 = 24 is divisible by 3. 1
Reason (R): The sum of three consecutive numbers equals 3 times the middle number.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The hailstone sequence starting from 6 reaches 1. 1
Reason (R): Kaprekar's routine on a 4-digit number leads to 6174.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Using each of the digits $9, 4, 8, 2$ once, make the largest 4-digit number divisible by 2. 2

12. A two-digit number added to its reverse gives 99 and subtracted from it gives 27. Find the number. 2

13. Show that $n(n+1)$ is even, and verify for $n = 50$. 2

14. Using each of the digits $8, 5, 4, 1$ once, make the largest 4-digit number divisible by 9. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The number $8 * 1 * 5$ has the same digit $*$ in both blanks and is divisible by 9. Find all possible values of $*$. 3

16. Start with 1368 and apply Kaprekar's routine until 6174 appears. Show every step. 3

17. In how many ways can 36 be written as the sum of two or more consecutive positive whole numbers? List them. 3

OR

The number $2 * 7 * 3$ has the same digit $*$ in both blanks and is divisible by 9. Find all possible values of $*$. 3

18. Start with 1374 and apply Kaprekar's routine until 6174 appears. Show every step. 3

OR

In how many ways can 45 be written as the sum of two or more consecutive positive whole numbers? List them. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A ticket has the number 63*8, where * is a missing digit. Rakesh wants the number to be divisible by 3 or 9. 5
(i) What is the sum of the known digits?
(ii) For which digits * is the number divisible by 3?
(iii) For which digit * is it divisible by 9?
(iv) Write the number for part
(iii) and verify by finding its quotient when divided by 9.

20. Use reverse-and-add on 156 until a palindrome is formed. Show each step, state the number of steps and the final palindrome. 5

OR

Find all two-digit numbers that are exactly 8 times the sum of their digits. 5

END OF QUESTION PAPER — Total: 40 Marks